

L5 Planning Quiz ANS

1. What is the primary output of mission planning in an autonomous-driving system?

- A) A map-level route from the start to the destination
- B) A steering-angle command
- C) A set of detected objects
- D) A throttle and brake signal

ANS: A) A map-level route from the start to the destination

2. What is the primary output of behavior planning?

- A) A raw occupancy grid
- B) A selected high-level driving maneuver, such as stop, follow, or change lanes
- C) A sequence of actuator voltages
- D) A global HD map

ANS: B) A selected high-level driving maneuver, such as stop, follow, or change lanes

3. What does motion planning generate for the vehicle controller to track?

- A) A traffic-rule database
- B) A destination address
- C) A feasible reference trajectory
- D) A sensor-calibration table

ANS: C) A feasible reference trajectory

4. In route planning, the road network is commonly represented as which structure?

- A) An unweighted image
- B) A finite-state controller
- C) A speed profile
- D) A weighted graph

ANS: D) A weighted graph

5. What problem does route planning solve?

- A) Finding an optimal navigation path from start to goal
- B) Predicting the color of a traffic light
- C) Estimating tire pressure
- D) Selecting a camera exposure

ANS: A) Finding an optimal navigation path from start to goal

6. Which value does Dijkstra's algorithm use to prioritize a node?

- A) Only the estimated cost from that node to the goal
- B) The currently known cost from the start to that node
- C) The node's geographic latitude
- D) The number of outgoing edges only

ANS: B) The currently known cost from the start to that node

7. In A* search, what does the evaluation value normally combine?

- A) Vehicle speed and steering angle
- B) Road width and lane count
- C) Cost from the start and estimated cost to the goal
- D) Camera and LiDAR confidence

ANS: C) Cost from the start and estimated cost to the goal

8. What property must a heuristic have to be admissible for A* optimality?

- A) It must always equal zero
- B) It must overestimate the remaining cost
- C) It must ignore the goal
- D) It must never overestimate the true remaining cost

ANS: D) It must never overestimate the true remaining cost

9. Why can A* explore fewer positions than Dijkstra's algorithm?

- A) Its heuristic directs the search toward the target
- B) It ignores all edge costs
- C) It expands nodes randomly
- D) It searches only backward from the goal

ANS: A) Its heuristic directs the search toward the target

10. Which information should a behavior planner consider when choosing a driving maneuver?

- A) Only the destination coordinates
- B) Road rules plus static and dynamic objects
- C) Only the current steering angle
- D) Only the vehicle's battery level

ANS: B) Road rules plus static and dynamic objects

11. What does the 'Follow Leader' maneuver require the ego vehicle to do?

- A) Pass the leading vehicle immediately
- B) Stop regardless of traffic
- C) Match the leader's speed while maintaining a safe distance
- D) Ignore the leading vehicle

ANS: C) Match the leader's speed while maintaining a safe distance

12. In a finite-state-machine behavior planner, what normally triggers a transition between states?

- A) A randomly selected timer
- B) A software recompilation
- C) A change in map file format
- D) An input condition or event

ANS: D) An input condition or event

13. How does reinforcement-learning-based behavior planning learn a policy?

- A) Through trial and error to maximize cumulative reward
- B) By storing only hand-written traffic rules

- C) By applying Dijkstra's algorithm to every maneuver
- D) By eliminating all environmental feedback

ANS: A) Through trial and error to maximize cumulative reward

14. Why might an FSM use $\text{Ego.Velocity} \leq \text{StopThreshold}$ instead of $\text{Ego.Velocity} == 0$?

- A) To increase the maximum speed
- B) To improve robustness to state-estimation noise
- C) To disable stopping behavior
- D) To remove the need for sensors

ANS: B) To improve robustness to state-estimation noise

15. Why is there no direct transition from 'Decelerate to Stop' to 'Track Speed' at a stop-sign intersection?

- A) The vehicle must always turn left
- B) The intersection has no approaching zone
- C) The vehicle must first come to a complete stop
- D) The behavior planner cannot measure speed

ANS: C) The vehicle must first come to a complete stop

16. How does a hierarchical FSM organize multiple driving scenarios?

- A) It uses one state for all situations
- B) It removes all entry and exit transitions
- C) It replaces behavior planning with route planning
- D) It models each scenario as a super-state containing a scenario-specific lower-level FSM

ANS: D) It models each scenario as a super-state containing a scenario-specific lower-level FSM

17. Which is a major limitation of FSM-based behavior planning?

- A) State explosion as scenarios become numerous and complex
- B) Inability to represent any driving maneuver
- C) Requirement that every road be straight
- D) Inability to use logical conditions

ANS: A) State explosion as scenarios become numerous and complex

18. How is a trajectory distinguished from a geometric path?

- A) A trajectory contains no spatial information
- B) A trajectory is a path parameterized by time
- C) A trajectory is always a straight line
- D) A trajectory contains only map-level turns

ANS: B) A trajectory is a path parameterized by time

19. Into which two main tasks can motion planning be decomposed?

- A) Localization and mapping
- B) Detection and segmentation
- C) Path planning and speed-profile generation

D) Mission planning and sensor fusion

ANS: C) Path planning and speed-profile generation

20. What does an occupancy grid encode?

A) Only the vehicle's control gains

B) Only the destination and start points

C) Only dynamic-object identities

D) Occupied and free regions in a discretized environment

ANS: D) Occupied and free regions in a discretized environment

21. What is a key difference between RRT and PRM when start and goal locations change?

A) RRT typically grows a new graph, whereas PRM can reuse its roadmap

B) PRM cannot represent obstacles

C) RRT always reuses one fixed roadmap

D) PRM does not require path search

ANS: A) RRT typically grows a new graph, whereas PRM can reuse its roadmap

22. After constructing a probabilistic roadmap, which algorithms can be used to find a route through it?

A) PID or MPC

B) A* or Dijkstra's algorithm

C) K-means or PCA

D) Backpropagation or gradient descent

ANS: B) A* or Dijkstra's algorithm

23. What does it mean that many RRT and PRM variants are probabilistically complete?

A) They always return the shortest path

B) Their runtime is fixed and predictable

C) If a path exists, it will be found in finite time, although the time may be very long

D) They never require collision checking

ANS: C) If a path exists, it will be found in finite time, although the time may be very long

24. Why is path smoothing often applied after sampling-based planning?

A) To add more collisions to the route

B) To convert the route into a global map

C) To increase unnecessary turns

D) To reduce path length and produce a smoother, easier-to-track path

ANS: D) To reduce path length and produce a smoother, easier-to-track path

25. What is the function of speed-profile generation?

A) Assigning a speed value to every point along a path

B) Selecting the destination city

C) Classifying road signs

D) Building the occupancy grid

ANS: A) Assigning a speed value to every point along a path

26. What characterizes a linear-ramp speed profile?

A) Constant speed at all times

B) Constant acceleration until the reference speed is reached

C) Random acceleration and braking

D) A mandatory full stop at every path point

ANS: B) Constant acceleration until the reference speed is reached

27. Which profile is described as constant-zero-constant deceleration and can be used to stop at a red light?

A) A circular profile

B) A linear-ramp profile

C) A trapezoidal profile

D) A probabilistic roadmap

ANS: C) A trapezoidal profile

28. Why is 'avoid collisions at all costs' not a practical autonomous-driving objective?

A) It requires all vehicles to drive faster

B) It eliminates the need for safety rules

C) It guarantees the AV is responsible for every crash

D) It can make the system so conservative that it becomes useless

ANS: D) It can make the system so conservative that it becomes useless

29. What is the central safety goal of Responsibility-Sensitive Safety (RSS)?

A) The AV should not cause accidents and should properly respond to others' mistakes

B) The AV should never be involved in any collision under any circumstances

C) The AV should always claim right-of-way

D) The AV should ignore implicit traffic rules

ANS: A) The AV should not cause accidents and should properly respond to others' mistakes

30. In RSS, what is 'blame time'?

A) The moment an insurance claim is filed

B) The first moment when a situation becomes dangerous

C) The time at which route planning begins

D) The delay between two sensor frames

ANS: B) The first moment when a situation becomes dangerous